



EAST WEST UNIVERSITY

LAB REPORT

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Course Name: Electronic Circuits

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Submitted By:

Name	Student ID
Naheda Islam Ela	2024-3-60-080
Abdullah Al Junaid	2024-3-60-082
Jumana Haider	2024-3-60-594

Submitted To:

Dr. Md. Habibur Rahman

Adjunct faculty

Department of Computer Science and Engineering

East west University

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Experiment Title

Study the Performance of Half-Wave Rectifier with and without Capacitor Filter

Objectives

- To construct a half-wave rectifier circuit.
- To observe the input and output waveforms of a half-wave rectifier.
- To study the effect of a capacitor filter on the rectifier output.
- To compare the theoretical and practical average output voltage without a capacitor filter.
- To compare the theoretical and practical ripple factor with a capacitor filter.
- To measure the forward voltage drop of the diode.

Theory

A half-wave rectifier converts an alternating current (AC) into a pulsating direct current (DC) by allowing only one half-cycle of the AC signal to pass through the load. During the positive half-cycle, the diode is forward biased and conducts current. During the negative half-cycle, the diode is reverse biased and blocks current.

The output of a half-wave rectifier contains large ripples, making it unsuitable for most DC applications. A capacitor filter connected across the load stores charge during the conducting period and releases it during the non-conducting period. As a result, the output voltage becomes smoother and the ripple is reduced significantly.

$$V_{\text{avg}} = \frac{V_m}{\pi}$$

where,

V_m = Peak output voltage

The ripple factor for a half-wave rectifier with capacitor filter is

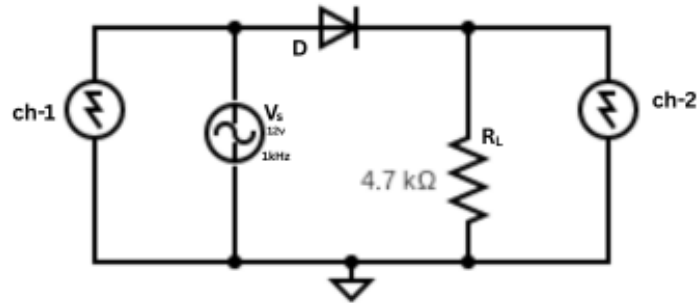
$$r = \frac{1}{2fR_L C}$$

where,

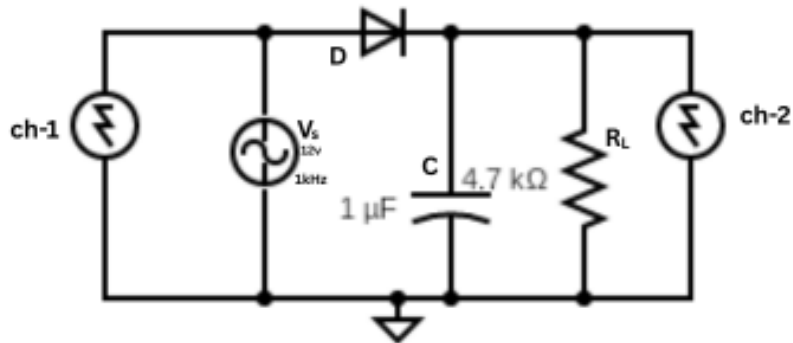
- f = Supply frequency
- R_L = Load resistance
- C = Filter capacitance

Working Circuit Diagram

Without Capacitor Filter



With Capacitor Filter

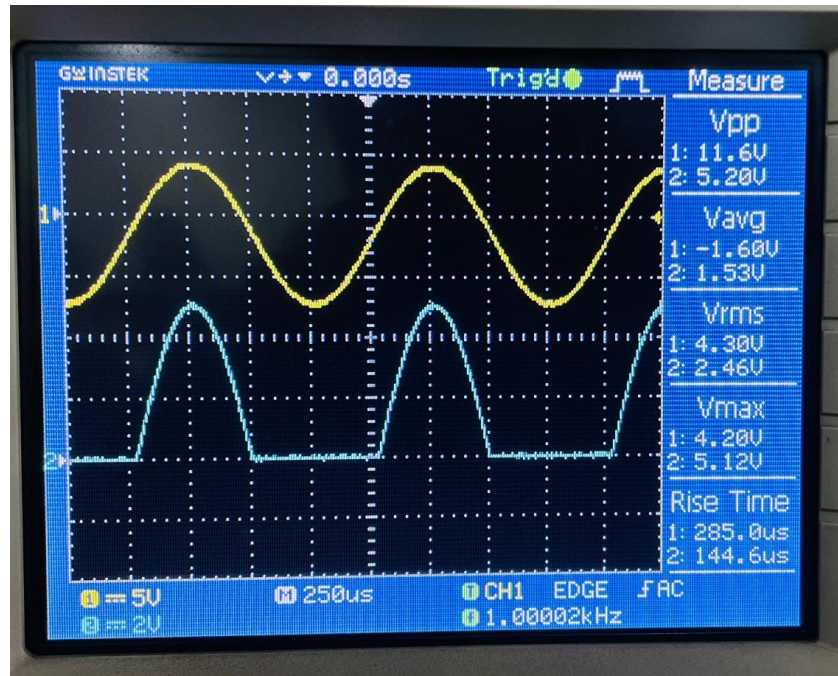


Components and Apparatus Required

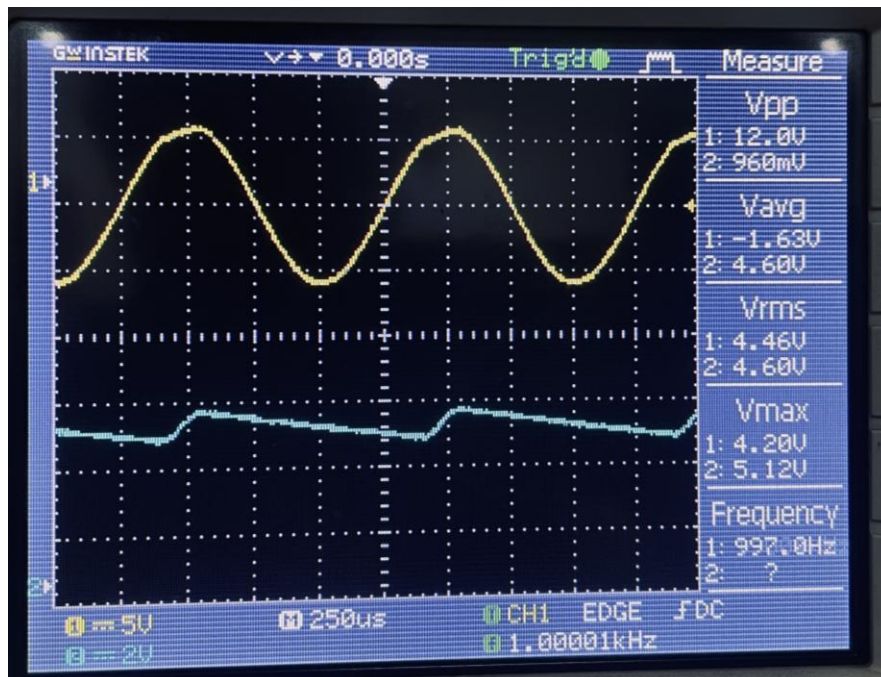
- Function Generator
- Oscilloscope
- Breadboard
- Resistor (4.7kΩ)
- Capacitor (1μF)
- Connecting Wires

Experimental Observations

Without Rectifier Filter:



With Rectifier Filter:



Calculation

Given,

Frequency, $f = 1 \text{ kHz}$

Load Resistance, $R_L = 4.7 \text{ k}\Omega$

Filer Capacitor, $C = 1 \text{ }\mu\text{F}$

A. Half-Wave Rectifier without Capacitor Filter

Practical Readings

Input Maximum Voltage,

$$V_{\max(\text{in})} = 4.20\text{V}$$

Output Maximum Voltage,

$$V_{\max(\text{out})} = 5.12\text{V}$$

Average Output Voltage,

$$V_{\text{avg}(\text{practical})} = 1.53\text{V}$$

Diode Forward Voltage Drop

$$V_D = V_{\max(\text{out})} - V_{\max(\text{in})}$$

$$= 5.12 - 4.20$$

$$V_D = 0.92\text{V}$$

Theoretical Average Output Voltage

$$V_{\text{avg}} = \frac{V_m}{\pi}$$

where, $V_m = 5.12\text{V}$

Therefore,

$$V_{\text{avg}} = \frac{5.12}{\pi} = 1.63\text{V}$$

Percentage Error

$$\text{Error} = \frac{1.63 - 1.53}{1.63} \times 100$$

$$= 6.13\%$$

B. Half-Wave Rectifier with Capacitor Filter

Practical Readings

Average Output Voltage,

$$V_{\text{avg(practical)}} = 4.60\text{V}$$

Ripple Peak-to-Peak Voltage,

$$V_{\text{pp}} = 0.96\text{V}$$

Practical Ripple Factor

Ripple Peak Voltage,

$$\begin{aligned} V_p &= \frac{V_{pp}}{2} \\ &= \frac{0.96}{2} \\ &= 0.48\text{V} \end{aligned}$$

Ripple RMS Voltage,

$$\begin{aligned} V_{\text{rms}} &= \frac{V_p}{\sqrt{3}} \\ &= \frac{0.48}{\sqrt{3}} \\ &= 0.277\text{V} \end{aligned}$$

Ripple Factor,

$$r_{\text{practical}} = \frac{V_{rms}}{V_{avg}}$$

$$= \frac{0.277}{4.60}$$

$$= 0.0060$$

$$= 6.0\%$$

Theoretical Ripple Factor

$$r_{\text{theoretical}} = \frac{1}{2fR_LC}$$

$$= \frac{1}{2 \times 1000 \times 4700 \times 1 \times 10^{-6}}$$

$$= \frac{1}{9.4}$$

$$= 0.106 = 10.6\%$$

Percentage Error

$$\text{Error} = \frac{0.106 - 0.060}{0.106} \times 100$$

$$= 43.4\%$$

Result

- The practical average output voltage of the half-wave rectifier without capacitor filter was 1.53 V, while the theoretical value was 1.63 V.
- The practical diode forward voltage drop was 0.92 V.
- With the capacitor filter, the practical ripple factor was 6.0%, whereas the theoretical ripple factor was 10.6%.
- The capacitor filter significantly increased the average DC output voltage and reduced the ripple content, resulting in a smoother DC output.

Discussion

In this experiment, the performance of a half-wave rectifier was studied with and without a capacitor filter. The input and output waveforms were observed using an oscilloscope, and the practical values were compared with the theoretical calculations.

For the half-wave rectifier without a capacitor filter, the output waveform consisted of only the positive half-cycles of the input AC signal. As a result, the output was a pulsating DC voltage with a low average value. The practical average output voltage was found to be slightly lower than the theoretical value. This difference occurred due to the forward voltage drop across the diode, internal resistance of the components, wiring losses, and small measurement errors in the oscilloscope.

After connecting the capacitor filter across the load, a significant improvement in the output waveform was observed. The capacitor charged quickly when the diode was forward biased and discharged slowly through the load resistor when the diode was reverse biased. This charging and discharging action reduced the ripple in the output voltage and produced a smoother DC output. The average output voltage also increased compared to the circuit without the capacitor filter.

The practical ripple factor was found to be lower than the theoretical value. This difference may be caused by component tolerances, the non-ideal characteristics of the diode and capacitor, oscilloscope measurement accuracy, and experimental limitations. In practical circuits, components do not behave exactly as assumed in theoretical calculations, so a small difference between theoretical and practical results is expected.

Overall, the experiment successfully demonstrated the working principle of a half-wave rectifier and the importance of using a capacitor filter. The capacitor filter effectively reduced the ripple content and improved the quality of the DC output voltage. The experimental results were reasonably close to the theoretical values, confirming the expected behavior of the rectifier circuit.